Robust Spatial–Spectral Block-Diagonal Structure Representation With Fuzzy Class Probability for Hyperspectral Image Classification

Yun Ding, Shaoming Pan, and Yanwen Chong

***Abstract*— Generally, the apparent spectral information of hyperspectral images (HSIs) is directly used to measure the similarity among HSI pixels in the feature space, but this process cannot reveal the inherent characteristics of HSI pixels. Moreover, constructing spatial–spectral block-diagonal subspace structure representations of intraclass land-cover samples remains a challenge for low-rank representation (LRR) in HSI classification. In this article, we propose two methods to reveal the complex intrinsic spatial–spectral features of HSIs using block-diagonal subspace structures, namely, the spatial–spectral block-diagonal structure representation with class probability (SSBDCP) and spatial-spectral block-diagonal structure representation with fuzzy class probability (SSBDFCP) methods, for HSI classification. First, the SSBDFCP and SSBDCP methods explore the structure similarity characteristics of the latent subspace to form a block-diagonal LRR (BDLRR) of intraclass pixels with class probability and fuzzy class probability (FCP) and suppress the interclass pixels' representations. Then, the spatial information is considered in the proposed methods to enhance spatial–spectral graph expression and capture more comprehensive information. Note that SSBDFCP can perform better than SSBDCP because the FCP considers the "typicalness" that a sample belongs to a specific category and utilizes complex intrinsic discriminative information based on the feedback of "weakly" supervised information. Moreover, the feedback information can remove the noise features around the pixels and take advantage of the benefits of true neighbors. The experimental results for the Indian Pines and Pavia University data sets show that the SSBDFCP and SSBDCP methods achieve better HSI classification results than other popular graph construction methods.**

***Index Terms*— Block-diagonal representation, fuzzy class probability (FCP), hyperspectral image (HSI), typicalness.**

I. Introduction

HYPERSPECTRAL images (HSIs) can be obtained with imaging spectrometers over hundreds of contiguous and narrow spectral bands to provide detailed information for land-cover classification [1], [2]. However, effective land-cover classification remains a key challenge due to the abundance of redundant spectral band information and limited sample sizes [3], [4]. Fortunately, graph-based representation learning algorithms have received extensive attention in HSI classification because of their convexity [5] and effectiveness in practice [6].

Graph-based learning algorithms have been applied for the classification of remote sensing data and can be broadly divided into two categories: unsupervised and supervised techniques. Unsupervised methods cannot utilize the labeled information to construct a graph. For example, the k-nearest neighbor (KNN) [7] and locally linear embedding (LLE) [8] methods are the two most popular graph construction methods. In contrast to unsupervised graph-based methods, supervised graph-based methods can use labeled information to construct better graphs for discriminative classification of unlabeled samples. This article focuses on supervised graph-based methods for HSI classification since they are typically functionally convex and have closed-form solutions [9]. Every sample is represented by a vertex in a weighted graph that measures the similarity between the samples. The key step in supervised graph-based methods is constructing better graphs to represent the original sample (labeled and unlabeled) structures. This process can be separated into two steps. First, a graph is constructed from the labeled and unlabeled samples. Then, the labeled information can be effectively propagated to the unlabeled samples through the graph [10]. In practice, the prediction of these unlabeled samples using a graph mainly benefits from the dependence on the labeled information. In other words, the unlabeled samples share the same structure or class as the corresponding labeled information based on the smoothness, cluster, and manifold assumptions among the samples [11]. The samples with the same manifold or cluster structure are likely to share the same class. Therefore, exploring the structural relationships between the unlabeled and labeled samples is critical in graph-based methods.

Traditional graph-based methods, such as the Gaussian kernel (GK) function [12] and nonnegative local linear reconstruction (LLR) [9] methods exploit the graph adjacency structure to measure the similarity of the samples. Recently, Ma *et al.* [8] proposed the local manifold learning (LML) method based on sparse graphs to capture local geometric properties for HSI classification. However, this method is not robust to the classification results because the neighboring points usually contain noise and unfavorable features.

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With the development of compressed sensing, series of sparse and low-rank graph methods have also received attention. Zhang *et al.* [13] proposed a spectral–spatial sparse graph for HSI clustering that considers spectral information and spatial information in the sparse subspace clustering (SSC) [14] to obtain adjacent coefficients and measure the similarity between the samples. Du *et al.* [15] proposed a sparse graph with total variation (TV) for HSI classification. Then, He *et al.* [16] proposed TV-regularized sparse and low-rank graphs for HSI restoration to remove mixed and sparse noise for improved restoration. In addition, some probabilistic graph techniques based on sparse representation (SR) have exhibited good performance in HSI classification. For example, Shao *et al.* [10] proposed a probabilistic class structure-regularized SR (PCSSR) graph method for HSI classification that included the class probability in the SR model to yield a discriminative graph. The problem of propagating labeled information to unlabeled samples via a PCSSR graph was explored. Next, the spatial and class structure-regularized SR (SCSSR) [17] graph for HSI classification was proposed. This method considered the detailed spatial information of an HSI based on a PCSSR graph. In addition, Yan *et al.* [18] proposed the class probability propagation of supervised information based on an SSC (CPPSSC) graph for the semisupervised clustering of HSI. The latent philosophy based on SR graphs is that each sample can be encoded as a sparse linear combination of the remaining samples by solving an l_1 optimization problem. In fact, SR graph methods generally benefit from their true neighbors and are usually sensitive to noise and parameter variance because of their local geometric properties.

To manage the local properties associated with improving graph representations between the samples, low-rank graph methods have been proposed; these methods can capture the global properties of samples from an entire data set [19]. Zhuang *et al.* [20] proposed a negative low-rank and sparse (NNLRS) graph for image classification, which can capture the global and local subspace structures. Moreover, this method is more robust to outliers to some extent. Fang *et al.* [21] proposed a robust subspace clustering method based on a nonnegative low-rank representation (NNLRR) graph. The method explicitly combined affinity matrix construction and subspace clustering into a unified step to obtain the overall optimum with the help of labeled information. In practice, NNLRR is an extension of NNLRS in graph construction. Neglecting the semantic information around HSI pixels, Wang *et al.* [22] proposed a self-supervised low-rank representation (SSLRR) graph for HSI classification that utilized LRR and spectral–spatial graph regularization to develop pixel–pixel constraints and remove noise. Ideally, the SR and LRR graph-based methods should be block-diagonal without noise, indicating that each sample in a data set can be ideally represented by a linear combination of a few points from the corresponding subspace. Consequently, Li *et al.* [23] proposed an LRR graph with a classwise block-diagonal structure for classification. Zhang *et al.* [24] developed the discriminative block-diagonal LRR (BDLRR) method, which improved the intraclass representation and enhanced the self-expressive power of the labeled and unlabeled samples. The ideal block-diagonal representation can capture the underlying semantic information around pixels by embedding the global semantic structural information [25] and thus semantic labels in representation learning, which is critical for the graph-based recognition problem. Moreover, the BDLRR model has the potential capability to address the out-of-sample problem outside training and testing samples, circumventing reimplementation of the time-consuming training model in the recognition problem [24]. However, the developed BDLRR model imposed an unsupervised constraint on the sparse property, which neglects the labeled information in sparse graph construction, and pure labeled information cannot propagate to unlabeled samples with the existing noise. Moreover, the latent class probabilities corresponding to specific classes cannot be considered in capturing the structure information in BDLRR. In fact, the latent class probabilities are favorable for exploring the latent structure representations of the subspace structures.

Overall, to make the proposed method more effective, the block-diagonal structure is adopted in our method. Adopting block-diagonal structure methods for HSI classification must solve three problems. One of the problems is that the learned intraclass representations are usually affected by noise. Another problem is that most discriminative graph construction methods lack the guidance of labeled information in exploring the structural relationships between the labeled and unlabeled samples. The last problem is that the latent class probabilities or memberships of the subspace structures are typically neglected. Motivated by the earlier discussion, in this article, we propose two discriminative graph construction methods called robust spatial–spectral block-diagonal structure representation with class probability (SSBDCP) and robust spatial–spectral block-diagonal structure representation with fuzzy class probability (SSBDFCP) for HSI classification. Specifically, class probability regularization is incorporated into the BDLRR model to guide graph construction with labeled information and explore the structural relationship between each sample and each class structure. In addition, fuzzy class probability (FCP) is further considered in the specific latent class probability relationship with "typicalness" since FCP can provide useful "weakly" supervised information to build a better affinity graph and facilitate label propagation. Then, the labeled sample representation for any unlabeled sample representation is used to remove the mixed noise around pixels. Finally, considering the rich spatial information of HSIs, the latent relationship between the spectral information and the spatial information of HSIs is determined to construct a discriminative spatial–spectral graph for HSI classification. The SSBDFCP model can be solved with the recently developed alternating direction with adaptive penalty [alternating direction method of multipliers (ADMM)] [26] algorithm. The experimental results for two hyperspectral data sets demonstrate the satisfactory performance of SSBDFCP. In summary, the key contributions of this article are as follows.

1) The class probability representation of the subspace structure is considered in the BDLRR model to guide graph construction for label propagation, which yields a discriminative graph-based representation.

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2) FCP regularization is explicitly integrated to effectively reveal the "typicalness" of the latent subspace structure from the surrounding neighbors and to provide a basis for the affinity graph by providing feedback via "weakly" supervised information.
3) Considering the abundant spatial information in HSIs, the latent structural relationship between the spectral information and the spatial information is developed for information integration.

The rest of this article is organized as follows. Related graph construction studies are briefly reviewed in Section II. Then, the proposed SSBDCP and SSBDFCP methods are presented in detail in Section III. Relevant solutions and discussion are provided in Section IV, as well as a convergence analysis of the SSBDFCP method. Section V reports the extensive experimental results and a parameter sensitivity analysis. Finally, the conclusion is drawn in Section VI.

II. Related Work

In this section, we briefly review some related works, including those focusing on the BDLRR [24] and PCSSR [17] methods. First, some notations used in this article are introduced. We denote matrices by bold uppercase letters, e.g., $\boldsymbol{X}$ and $\boldsymbol{Z}$, and denote column vectors by lowercase letters, e.g., x. Moreover, $\mathbf{I}$ represents the identity matrix, and x_{ij} represents the ith row and jth column element of matrix $\boldsymbol{X}$. For a matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N] \in \mathbb{R}^{\boldsymbol{d}\times N}$, the $l_{u,v}$ norm can be obtained via

$$\|\boldsymbol{X}\|_{u,v} = \left[\sum_{i=1}^{d}\left(\sum_{j=1}^{N}|\boldsymbol{x}_{ij}|^{u}\right)^{v/u}\right]^{1/v}. \tag{1}$$

The Frobenius norm can be obtained when $u = v = 2$, the $l_{2,1}$ norm is obtained when $v = 1$ and $u = 2$, and the sparse l_1 norm is obtained when $u = v = 1$. In essence, the Frobenius norm of matrix $\boldsymbol{X}$ can be formulated as an l_2 norm that can be rewritten as follows:

$$\|\boldsymbol{X}\|_F^2 = \sum_{i=1}^{d}\sum_{j=1}^{N}\boldsymbol{x}_{ij}^2 = \sum_{i=1}^{d}\|\boldsymbol{x}^i\|_2^2 = \mathrm{tr}(\boldsymbol{X}^T\boldsymbol{X}) = \mathrm{tr}(\boldsymbol{X}\boldsymbol{X}^T) \tag{2}$$

where $\boldsymbol{x}^i$ represents the ith row vector of $\boldsymbol{X}$. The $l_{2,1}$ norm of the matrix $\boldsymbol{X}$ can be formulated as follows:

$$\|\boldsymbol{X}\|_{2,1} = \sum_{i=1}^{d}\left[\sum_{j=1}^{N}\boldsymbol{x}_{ij}^2\right]^{1/2} = \sum_{i=1}^{d}\|\boldsymbol{x}^i\|_2 = 2\mathrm{tr}(\boldsymbol{X}^{\mathrm{T}}\boldsymbol{O}\boldsymbol{X}) \tag{3}$$

where $\boldsymbol{O}$ is a diagonal matrix with $O_{ii} = 1/2\|\boldsymbol{x}^i\|_2$, $i = 1, 2, \ldots, d$ [27], [28]. In practice, $\|\boldsymbol{x}^i\|_2$ can be equal to zero when $\boldsymbol{x}^i$ is the zero-row vector. Therefore, in such a case, we can set $O_{ii} = 1/(2\|\boldsymbol{x}^i\|_2 + \tau)$ via regularization [29] to guarantee the practical significance of the result, where τ is a small value. Consequently, $2\|\boldsymbol{x}^i\|_2$ can be approximated by $(2\|\boldsymbol{x}^i\|_2 + \tau)$ to some extent.

$\|\boldsymbol{X}\|_*$ is the nuclear norm of matrix $\boldsymbol{X}$, which is the sum of the singular values of matrix $\boldsymbol{X}$, i.e., $\|\boldsymbol{X}\|_* = \sum_i |\sigma_i|$. $\boldsymbol{X}^T$ denotes the transpose matrix of $\boldsymbol{X}$. 0_{mn} denotes an all-zeros matrix of size of $m \times n$, and the all-one vector denotes $\mathbf{1}_N = \underbrace{[1, \ldots, 1]}_{N}^{T}$.

A. BDLRR

The objective of the BDLRR [24] is to collaboratively learn the appropriate block-diagonal representations of labeled and unlabeled samples by jointly considering the incoherence of extraclass sample representations and enhancing the coherence of intraclass data representations. We consider the labeled data matrix $\boldsymbol{X}_{\mathrm{tr}} \in \mathbb{R}^{d\times l}$ and the unlabeled data matrix $\boldsymbol{X}_{\mathrm{tt}} \in \mathbb{R}^{d\times u}$ with dimension d, where l represents the number of the labeled samples and u represents the number of the unlabeled samples. The total number of samples is $N = l + u$. Both the labeled and unlabeled samples can be expressed as $\boldsymbol{X} = [\boldsymbol{X}_{\mathrm{tr}}, \boldsymbol{X}_{\mathrm{tt}}]$.

To obtain the ideal block-diagonal structure of the labeled and unlabeled samples together, the off-block-diagonal components can be naturally assumed to be as small as possible to suppress the extra-class representation. According to the self-expressiveness property [14], the objective function of the BDLRR algorithm can be formulated as follows:

$$\begin{aligned}&\min_{\boldsymbol{Z},\boldsymbol{E}} \|\boldsymbol{Z}\|_* + \lambda_1\|\tilde{\boldsymbol{A}}\odot\boldsymbol{Z}\|_F^2 + \lambda_2\|\boldsymbol{D}\odot\boldsymbol{Z}\|_1 + \lambda_3\|\boldsymbol{E}\|_{2,1}\\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{X}_{\mathrm{tr}}\boldsymbol{Z} + \boldsymbol{E}\end{aligned} \tag{4}$$

where λ_1, λ_2, and λ_3 are the balance parameters. Additionally, $\tilde{\boldsymbol{A}} = [\bar{\boldsymbol{A}}, \mathbf{1}_l\mathbf{1}_{N-l}^T]$ and $\bar{\boldsymbol{A}} = \mathbf{1}_l\mathbf{1}_l^T - \boldsymbol{M}$ are the constraints for minimizing the off-block-diagonal entries, and

$$\boldsymbol{M} = \begin{bmatrix}\mathbf{1}_{l1}\mathbf{1}_{l1}^T & \cdots & 0\\ \vdots & \ddots & \vdots\\ 0 & \cdots & \mathbf{1}_{lC}\mathbf{1}_{lC}^T\end{bmatrix}.$$

Moreover, $d_{ij} \in \boldsymbol{D}$ is a similarity measure of the Euclidean distance between all samples.

B. Probabilistic Class Structure Regularized Sparse Representation

Most representation learning [30], [31] methods fail to consider the class structure in the correlation of HSIs. To learn a discriminative representation between the labeled samples and unlabeled samples, Shao *et al.* [10] presented a discriminative (PCSSR) approach by utilizing a small number of labeled samples for HSI classification. The PCSSR method aims to effectively propagate labeled information to unlabeled samples through the PCSSR. The PCSSR model solves the following problem:

$$\begin{aligned}&\min_{\boldsymbol{Z}} \frac{1}{2}\|\boldsymbol{X} - \boldsymbol{X}\boldsymbol{Z}\|_{\mathrm{F}}^2 + \lambda_1\|\boldsymbol{Z}\|_1 + \lambda_2\|\hat{\Theta}\odot\boldsymbol{Z}\|_1\\ &\text{s.t. } \mathrm{diag}(\boldsymbol{Z}) = 0, \boldsymbol{Z} \geq 0\end{aligned} \tag{5}$$

where $\hat{\Theta}_{ij} = (1/2)\|\hat{\boldsymbol{\theta}}_i - \hat{\boldsymbol{\theta}}_j\|_2^2$ denotes the similarity measure of the class distribution between samples $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$, $\hat{\boldsymbol{\theta}}_i \in \mathbb{R}^{1\times C}$ is a class distribution vector of sample $\boldsymbol{x}_i$ corresponding to all classes, and $\odot$ indicates the Hadamard product (which is an elementwise product). The class distribution of labeled samples $\boldsymbol{X}_{\mathrm{tr}} \in \mathbb{R}^{l\times C}$ can be directly obtained by *a priori* knowledge. The estimated class probability of each unlabeled sample

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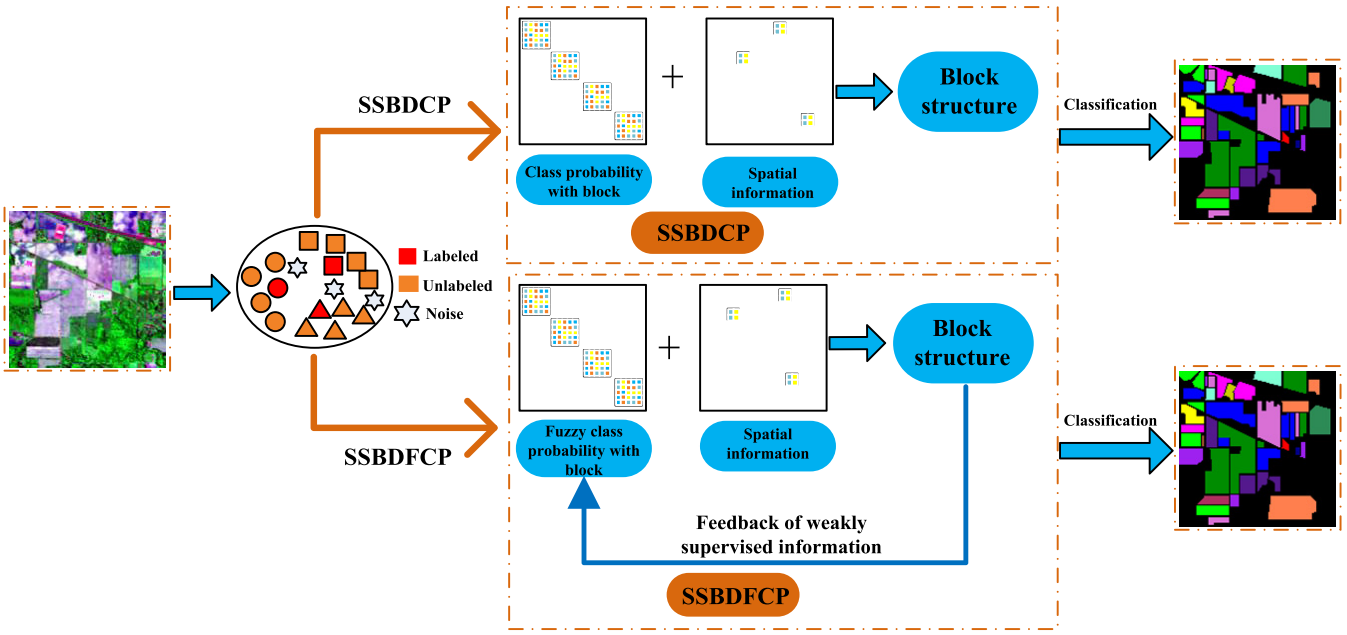


Fig. 1. Flowchart of the proposed SSBDCP and SSBDFCP methods for HSI classification.

$\boldsymbol{x}_k \in \boldsymbol{X}_{\text{tt}}$ can be encoded as a sparse linear combination of all labeled samples

$$\boldsymbol{a} = \arg\min \frac{1}{2}\|\boldsymbol{x}_k - \boldsymbol{X}_{\text{tr}}\boldsymbol{a}\|_F^2 + \lambda\|\boldsymbol{a}\|_1 \quad \text{s.t. } \boldsymbol{a} \geq 0. \tag{6}$$

The representation coefficient $\boldsymbol{a} \in \mathbb{R}^{l\times 1}$ indicates the similarity between the unlabeled sample $\boldsymbol{x}_k$ and labeled samples. Then, the class probability of each unlabeled sample can be denoted as follows:

$$\hat{\boldsymbol{\theta}}_k = \boldsymbol{a}^T \boldsymbol{Y} \tag{7}$$

where $\boldsymbol{Y} \in \mathbb{R}^{l\times C}$ is a label indicator binary matrix, i.e., $\boldsymbol{Y} \in \{0, 1\}$, $\boldsymbol{Y1} = \boldsymbol{1}$. All unlabeled samples in the class distribution form a class probability matrix $\hat{\boldsymbol{\theta}}_u \in \mathbb{R}^{u\times C}$. As a result, the class probability of all samples is $\hat{\boldsymbol{\theta}} = \begin{bmatrix} \boldsymbol{Y} \\ \hat{\boldsymbol{\theta}}_u \end{bmatrix} \in \mathbb{R}^{N\times C}$.

III. Proposed Methods

In this section, the SSBDCP and SSBDFCP methods are introduced to further enforce the intraclass representation for compactness in HSI classification to capture the latent subspace structure representation and yield a more discriminative block representation. In practice, SSBDFCP is the extension of SSBDCP in latent subspace structure representation and further facilitates the intraclass representation with weakly supervised information feedback from the representation vectors. The flowchart of the proposed algorithms is shown in Fig. 1.

A. BDLRR With Class Probability and Fuzzy Class Probability

For HSI data set $\boldsymbol{X} = [\boldsymbol{X}_1, \ldots, \boldsymbol{X}_{\text{C}}] \in \mathbb{R}^{d\times N}$, d denotes the number of bands, N is the number of HSI pixels, and C represents the number of land-cover types. Each sample from a union of multiple subspaces can be effectively represented by a linear combination of the other samples from its own subspace according to the self-expressiveness property [32].

The BDLRR model imposes the sparse constraint $\|\boldsymbol{D} \odot \boldsymbol{Z}\|_1$ and low-rank constraint $\|\boldsymbol{Z}\|_*$ to yield a block-diagonal structure representation. The objective of the Euclidean distance matrix $\boldsymbol{D}$ is to achieve a representation $\boldsymbol{Z}$ that is sparse with the help of the l_1 norm. However, this Euclidean distance matrix $\boldsymbol{D}$ is an unsupervised graph constraint that only reveals the apparent features that exist on the surface of samples, and the internal attribute features and latent subspace structure cannot be explored with this approach. Moreover, this matrix cannot effectively guide graph construction. Therefore, to address the two constraints, the class probability $\hat{\Theta}$ is introduced to reveal the block-diagonal structure representation, which can be formulated as follows:

$$\begin{aligned} &\min_{\mathbf{Z},\mathbf{E}} \|\mathbf{Z}\|_* + \lambda_1\|\tilde{\mathbf{A}} \odot \mathbf{Z}\|_F^2 + \lambda_2\|\hat{\Theta} \odot \mathbf{Z}\|_1 + \lambda_3\|\mathbf{E}\|_{2,1} \\ &\text{s.t. } \mathbf{X} = \mathbf{X}_{\text{tr}}\mathbf{Z} + \mathbf{E} \end{aligned} \tag{8}$$

where $\hat{\Theta} \in \mathbb{R}^{l\times N}$ represents the latent subspace structure similarity of the class probability obtained for labeled samples and all samples, i.e., $\hat{\Theta}_{ij} = (1/2)\|\hat{\boldsymbol{\theta}}_i - \hat{\boldsymbol{\theta}}_j\|_2^2$. The limitation of apparent feature constraint $\boldsymbol{D}$ is addressed by the class probability, which can effectively reveal the subspace structure information based on the class distribution of all samples. Additionally, this approach can effectively guide the intraclass representation with the help of labeled information.

However, considering the class probability limitation, we propose the FCP to further enhance the graph expression. First, from (6), the class probability of each sample in each class is given a specific value, and the class probability of a sample corresponding to its own true class will be small

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because the class probability of a sample must sum to one for mathematical tractability. Furthermore, we emphasize the importance of a sample belonging to its own true class. In other words, the class probability lacks "typicalness" and is given equal importance in determining the class distribution of the sample to be classified. Second, the computational complexity of the class probability is high and the run time is long for a large number of samples because of the computational expense associated with the l_1 norm [33]. Third, the class probability is incorporated into model (5) to capture the intrinsic structure in a transductive manner because the feedback of weakly supervised information related to the latent subspace structure is lacking. Therefore, we propose the FCP to further specify the "typicalness" that a sample belongs to the corresponding class and produce feedback information for better guidance.

Given a small quantity of labeled information, the FCP can explore the memberships of unlabeled samples within the corresponding category. The FCPs or memberships of unlabeled samples can be obtained by representation vectors [34] based on the following problem:

$$\bar{\boldsymbol{\theta}}_j = \frac{(1/\|(\boldsymbol{Z}_u)_j - (\boldsymbol{Z}_l)_i\|^2)}{\sum_{i=1}^{K}(1/\|(\boldsymbol{Z}_u)_j - (\boldsymbol{Z}_l)_i\|^2)}\boldsymbol{Y}_{K\times C}, \quad i = 1\ldots K \tag{9}$$

where $\boldsymbol{Z}_u \in \mathbb{R}^{u\times l}$ denotes the representation coefficient vectors of the unlabeled samples, $\boldsymbol{Z}_l \in \mathbb{R}^{l\times l}$ denotes the representation coefficient vectors of the labeled samples, $(\boldsymbol{Z}_u)_j$ and $(\boldsymbol{Z}_l)_i$ are the jth row coefficient vector of $\boldsymbol{Z}_u \in \mathbb{R}^{u\times l}$ and the ith row coefficient vector of $\boldsymbol{Z}_l \in \mathbb{R}^{l\times l}$, respectively. $\bar{\boldsymbol{\theta}}_j \in \mathbb{R}^{1\times C}$ represents the memberships of x_j with corresponding C classes. K is the number of representation coefficient vectors that labeled samples contribute to the unlabeled sample x_j, assuming that more similar samples correspond to more similar representation coefficient vectors. $\boldsymbol{Y} \in \mathbb{R}^{l\times C}, \boldsymbol{Y}\mathbf{1} = \mathbf{1}$, is the given label indicator binary matrix, where $\boldsymbol{Y}_{K\times C} \in \mathbb{R}^{K\times C}$ is the label indicator matrix of the K nearest labeled samples based on the minimum Euclidean distance between the representation coefficient vectors of labeled samples and the representation coefficient vector of an unlabeled sample. In practice, the label indicator matrix $\boldsymbol{Y}_{K\times C} \in \mathbb{R}^{K\times C}$ is obtained based on representation coefficient vectors between the labeled samples and unlabeled samples, which can be conducive to removing noise that exists in the surrounding pixels by selecting K nearest representation coefficient vectors of labeled samples for the unlabeled samples. According to (9), the memberships of unlabeled samples can be obtained, i.e., $\bar{\boldsymbol{\theta}}_{\text{tt}} \in \mathbb{R}^{u\times C}$. This relation demonstrates the local similarity between the unlabeled and labeled samples because, ideally, each unlabeled sample can be represented by the corresponding labeled samples from the same class. In other words, $\bar{\boldsymbol{\theta}}_{\text{tt}} \in \mathbb{R}^{u\times C}$ indicates that each unlabeled sample has a certain degree of similarity with the true discovered neighbors of labeled samples [35]. Therefore, the propagation of information from labeled samples to unlabeled samples occurs via the true discovered neighbors. A graphic interpretation of calculating the FCP of an unlabeled sample is shown in Fig. 2.

Finally, the FCP can be denoted as $\boldsymbol{\theta} = \begin{bmatrix} \mathbf{Y} \\ \bar{\boldsymbol{\theta}}_{\text{tt}} \end{bmatrix} \in \mathbb{R}^{N\times C}, \boldsymbol{\theta}\mathbf{1} = \mathbf{1}$, and this calculation benefits from information

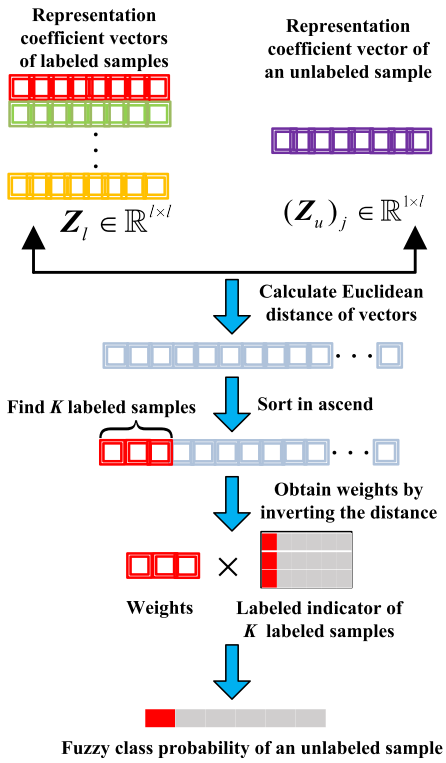


Fig. 2. Flowchart for calculating the FCP of an unlabeled sample.

from true neighbors. As shown in the FCP matrix $\boldsymbol{\theta}$, similar samples have high probabilities of being in the corresponding class, while dissimilar samples have a lower same-class probability. In this article, we simply define the similarity measurement of the FCP between any two samples as $\Theta_{ij} = (1/2)\|\boldsymbol{\theta}_i - \boldsymbol{\theta}_j\|_2^2$ imposed on the representation coefficient $\boldsymbol{Z}$. If the similarity degree is high between any two samples, the similarity degree of the representation coefficient $\boldsymbol{Z}$ will be high and the FCP will be small, and vice versa. Therefore, the block-diagonal structure of the FCP model can be formulated as follows:

$$\begin{aligned} &\min_{\mathbf{Z},\mathbf{E}} \|\mathbf{Z}\|_* + \lambda_1\|\tilde{\mathbf{A}} \odot \mathbf{Z}\|_F^2 + \lambda_2\|\Theta \odot \mathbf{Z}\|_1 + \lambda_3\|\mathbf{E}\|_{2,1} \\ &\text{s.t. } \mathbf{X} = \mathbf{X}_{\text{tr}}\mathbf{Z} + \mathbf{E} \end{aligned} \tag{10}$$

where the constraint $\|\Theta \odot \boldsymbol{Z}\|_1$ will be increasingly sparse as the number of labeled samples increases to improve the graph structure. Moreover, the feedback information provided by the representation coefficient vectors of the K nearest labeled samples can be obtained for better guidance.

B. Formulation of SSBDCP and SSBDFCP

Spatial information plays a vital role in the extraction of HSI information for land-cover classification. This unique spatial information must be utilized to provide features for graph guidance. We utilize a local neighbor window with a size of $s \times s$ to define the spatial information with respect to a set of HSI pixels. For each center pixel $\boldsymbol{x}_i$, the spatial neighborhood can be defined as $N_s(x_i)$ within the $s \times s$ local window. The spatial neighbor relationship between the labeled samples and

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the total samples is defined as follows:

$$\boldsymbol{S}_{ij} = \begin{cases} 1 & \text{if } \boldsymbol{x}_i \in N_s(\boldsymbol{x}_j) \text{ or } \boldsymbol{x}_j \in N_s(\boldsymbol{x}_i) \\ 0 & \text{otherwise.} \end{cases} \tag{11}$$

As shown in (11), the spatial information shared between the labeled and total samples has an approximate feature structure because the spatially close samples are more likely to be similar or to be represented by intraclass samples. Specifically, $\boldsymbol{S} \in \{0, 1\}$ is a binary matrix in which the representation of spatially similar samples within a spatial window is 1 and the representation of spatially dissimilar samples is 0. The structural combination of the spectral block-diagonal representation coefficient $\boldsymbol{Z} \in \mathbb{R}^{l\times N}$ and spatially neighbor relationship $\boldsymbol{S} \in \mathbb{R}^{l\times N}$ can be theoretically guaranteed [36]. The relationship between the spectral information and the spatial information is obtained. In addition, the robust SSBDFCP model for HSI classification is formulated as follows:

$$\begin{aligned} \min_{\mathbf{Z},\mathbf{E},\mathbf{W}} \ & \|\mathbf{Z}\|_* + \lambda_1\|\tilde{\mathbf{A}} \odot \mathbf{Z}\|_F^2 + \lambda_2\|\Theta \odot \mathbf{Z}\|_1 \\ & + \lambda_3\|\mathbf{Z} - \mathbf{S}\|_F^2 + \lambda_4\|\mathbf{E}\|_{2,1} \\ \text{s.t. } & \mathbf{X} = \mathbf{X}_{\text{tr}}\mathbf{Z} + \mathbf{E}. \end{aligned} \tag{12}$$

Moreover, the SSBDCP model can be formulated as follows:

$$\begin{aligned} \min_{\mathbf{Z},\mathbf{E}} \ & \|\mathbf{Z}\|_* + \lambda_1\|\tilde{\mathbf{A}} \odot \mathbf{Z}\|_F^2 + \lambda_2\|\hat{\Theta} \odot \mathbf{Z}\|_1 \\ & + \lambda_3\|\mathbf{Z} - \mathbf{S}\|_F^2 + \lambda_4\|\mathbf{E}\|_{2,1} \\ \text{s.t. } & \mathbf{X} = \mathbf{X}_{\text{tr}}\mathbf{Z} + \mathbf{E} \end{aligned} \tag{13}$$

where the constraint $\|\mathbf{Z} - \mathbf{S}\|_F^2$ facilitates the connectivity among intraclass samples and sets the extraclass representation to zero. The presented SSBDFCP algorithm is applied to an HSI. According to (13), the block-diagonal low-rank structure with FCP $\|\mathbf{Z}\|_* + \lambda_1\|\tilde{\mathbf{A}} \odot \mathbf{Z}\|_F^2 + \lambda_2\|\Theta \odot \mathbf{Z}\|_1$ aims to capture the full range of spectral information based on the BDLRR of the associated samples. Moreover, FCP is incorporated to further measure the local similarity representation of intraclass samples. Finally, the structure combination between the spectral features $\boldsymbol{Z}$ and the spatial features $\boldsymbol{S}$ yields the expression $\|\mathbf{Z} - \mathbf{S}\|_F^2$, which facilitates the spatial structure representation among similar samples and improves the spectral–spatial graph expression.

IV. Optimization and Analysis of SSBDFCP

A. Optimization

In this section, we briefly introduce the ADMM [26], [37] to solve the SSBDFCP algorithm, which can separate this problem into several subproblems based on the closed-form solutions. Then, the SSBDFCP problem can be rewritten as

$$\begin{aligned} \min_{\mathbf{Z},\mathbf{E},\mathbf{P},\mathbf{Q}} \ & \|\mathbf{P}\|_* + \lambda_1/2\|\tilde{\mathbf{A}} \odot \mathbf{Z}\|_F^2 + \lambda_2\|\Theta \odot \mathbf{Q}\|_1 \\ & + \lambda_3/2\|\mathbf{Z} - \mathbf{S}\|_F^2 + \lambda_4\|\mathbf{E}\|_{2,1} \\ \text{s.t. } & \mathbf{X} = \mathbf{X}_{\text{tr}}\mathbf{Z} + \mathbf{E}, \mathbf{P} = \mathbf{Z}, \mathbf{Q} = \mathbf{Z}. \end{aligned} \tag{14}$$

Here, the augmented Lagrangian function of the problem is formulated as follows:

$$\begin{aligned} & L(\mathbf{Z}, \mathbf{E}, \mathbf{P}, \mathbf{Q}) \\ & = \|\mathbf{P}\|_* + \frac{\lambda_1}{2}\|\tilde{\mathbf{A}} \odot \mathbf{Z}\|_F^2 + \lambda_2\|\Theta \odot \mathbf{Q}\|_1 \\ & \quad + \frac{\lambda_3}{2}\|\mathbf{Z} - \mathbf{S}\|_F^2 + \lambda_4\|\mathbf{E}\|_{2,1} + \langle\hat{\mathbf{Y}}_1, \mathbf{X} - \mathbf{X}_{\text{tr}}\mathbf{Z} - \mathbf{E}\rangle \\ & \quad + \langle\hat{\mathbf{Y}}_2, \mathbf{P} - \mathbf{Z}\rangle + \langle\hat{\mathbf{Y}}_3, \mathbf{Q} - \mathbf{Z}\rangle \\ & \quad + \frac{\mu}{2}\left(\|\mathbf{P} - \mathbf{Z}\|_F^2 + \|\mathbf{Q} - \mathbf{Z}\|_F^2 + \|\mathbf{X} - \mathbf{X}_{\text{tr}}\mathbf{Z} - \mathbf{E}\|_F^2\right) \end{aligned} \tag{15}$$

where $\text{tr}(\cdot)$ is a trace operator. $\hat{\mathbf{Y}}_1$, $\hat{\mathbf{Y}}_2$, and $\hat{\mathbf{Y}}_3$ are the Lagrangian multipliers. In addition, $\mu > 0$ is the penalty parameter. The function $\boldsymbol{L}$ is minimized with respect to one variable and one coordinate direction at each iteration, and the other variables are fixed. Here, the ADMM algorithm updates the variables $\mathbf{Z}$, $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{E}$ and the Lagrangian multipliers $\hat{\mathbf{Y}}_1$, $\hat{\mathbf{Y}}_2$, and $\hat{\mathbf{Y}}_3$ via the detailed procedures given next.

1) $\boldsymbol{Z}$ is updated by solving the following problem, while all other variables are fixed:

$$\begin{aligned} L_{\mathbf{Z}} = & \min_{\mathbf{Z}} \frac{\lambda_1}{2}\|\tilde{\mathbf{A}} \odot \mathbf{Z}\|_F^2 + \frac{\lambda_3}{2}\|\mathbf{Z} - \mathbf{S}\|_F^2 \\ & + \left\langle\hat{\mathbf{Y}}_1^t, \mathbf{X} - \mathbf{X}_{\text{tr}}\mathbf{Z} - \mathbf{E}^t\right\rangle + \left\langle\hat{\mathbf{Y}}_2^t, \mathbf{P}^{t+1} - \mathbf{Z}\right\rangle \\ & + \left\langle\hat{\mathbf{Y}}_3^t, \mathbf{Q}^t - \mathbf{Z}\right\rangle \\ & + \frac{\mu^t}{2}\left(\|\mathbf{P}^{t+1} - \mathbf{Z}\|_F^2 + \|\mathbf{Q}^t - \mathbf{Z}\|_F^2 \right. \\ & \qquad \left. + \|\mathbf{X} - \mathbf{X}_{\text{tr}}\mathbf{Z} - \mathbf{E}^t\|_F^2\right) \\ = & \frac{\lambda_1}{2}\|\tilde{\mathbf{A}} \odot \mathbf{Z}\|_F^2 + \frac{\lambda_3}{2}\|\mathbf{Z} - \mathbf{S}\|_F^2 \\ & + \frac{\mu^t}{2}\left(\left\|\mathbf{X} - \mathbf{X}_{\text{tr}}\mathbf{Z} - \mathbf{E}^t + \frac{\hat{\mathbf{Y}}_1^t}{\mu^t}\right\|_F^2 \right. \\ & \qquad + \left\|\mathbf{P}^{t+1} - \mathbf{Z} + \frac{\hat{\mathbf{Y}}_2^t}{\mu^t}\right\|_F^2 \\ & \qquad \left. + \left\|\mathbf{Q}^t - \mathbf{Z} + \frac{\hat{\mathbf{Y}}_3^t}{\mu^t}\right\|_F^2\right) \end{aligned} \tag{16}$$

which is equivalent to the following equation:

$$\begin{aligned} L_{\mathbf{Z}} = & \min_{\mathbf{Z}} \frac{\lambda_1}{2}\|\ \mathbf{Z} - \bar{\mathbf{R}}\|_F^2 + \frac{\lambda_3}{2}\|\mathbf{Z} - \mathbf{S}\|_F^2 \\ & + \frac{\mu^t}{2}\left(\left\|\mathbf{X} - \mathbf{X}_{\text{tr}}\mathbf{Z} - \mathbf{E}^t + \frac{\hat{\mathbf{Y}}_1^t}{\mu^t}\right\|_F^2 \right. \\ & \qquad + \left\|\mathbf{P}^{t+1} - \mathbf{Z} + \frac{\hat{\mathbf{Y}}_2^t}{\mu^t}\right\|_F^2 \\ & \qquad \left. + \left\|\mathbf{Q}^t - \mathbf{Z} + \frac{\hat{\mathbf{Y}}_3^t}{\mu^t}\right\|_F^2\right) \end{aligned} \tag{17}$$

where $\bar{\boldsymbol{R}} = [\boldsymbol{M}, \mathbf{0}_{l(N-l)}] \odot \boldsymbol{Z}^t$. Equation (17) has an optimal closed-form solution that can be obtained by calculating the derivative of $L_{\boldsymbol{Z}}$ with respect to $\boldsymbol{Z}$ and

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setting it to zero as follows:

$$\boldsymbol{Z}^{t+1} = \left[\left(2+\frac{\lambda_1}{\mu^t}+\frac{\lambda_3}{\mu^t}\right)\boldsymbol{I}+\boldsymbol{X}_{tr}^T\boldsymbol{X}_{tr}\right]^{-1} \times\left(\frac{\lambda_1}{\mu^t}\bar{\boldsymbol{R}}+\boldsymbol{X}_{tr}^T\hat{\boldsymbol{R}}_1+\hat{\boldsymbol{R}}_2+\hat{\boldsymbol{R}}_3+\frac{\lambda_3}{\mu^t}\boldsymbol{S}\right) \quad (18)$$

where $\hat{\boldsymbol{R}}_1 = \boldsymbol{X}-\boldsymbol{E}^t+(\hat{\boldsymbol{Y}}_1^t/\mu^t)$, $\hat{\boldsymbol{R}}_2 = \boldsymbol{P}^{t+1}+(\hat{\boldsymbol{Y}}_2^t/\mu^t)$, and $\hat{\boldsymbol{R}}_3 = \boldsymbol{Q}^{t+1}+(\hat{\boldsymbol{Y}}_3^t/\mu^t)$.

2) $\boldsymbol{P}$ is updated by converting the objective function [see (15)] into a new function with respect to $\boldsymbol{P}$, while all other variables are fixed

$$\begin{aligned}\mathbf{P}^{t+1} &= \arg\min_{\mathbf{P}} \|\mathbf{P}\|_*+\left\langle\hat{\mathbf{Y}}_2^t,\mathbf{P}-\mathbf{Z}^t\right\rangle+\frac{\mu^t}{2}\|\mathbf{P}-\mathbf{Z}^t\|_F^2\\ &= \|\mathbf{P}\|_*+\frac{\mu^t}{2}\|\mathbf{P}-\left(\mathbf{Z}^t-\frac{\hat{\mathbf{Y}}_2^t}{\mu^t}\right)\|_F^2. \end{aligned} \quad (19)$$

Equation (19) can be solved in the closed form using the singular value thresholding operator [38], which is $\mathbf{P}^{t+1} = \Gamma_{(\mu^t)^{-1}}(\mathbf{Z}^t-\hat{\mathbf{Y}}_2^t/\mu^t) = \mathbf{U}\boldsymbol{\Sigma}\ \mathbf{V}^{\mathrm{T}}$. Σ is a singular value of $\mathbf{Z}^t-\hat{\mathbf{Y}}_2^t/\mu^t$, which can be obtained by the soft-thresholding operator $\hat{S}_{1/\mu^t}(\mathbf{Z}^t-\hat{\mathbf{Y}}_2^t/\mu^t)$ [39].

3) $\mathbf{Q}$ is updated by solving the following problem [see (21)] when all other variables are fixed:

$$\begin{aligned}L_{\mathbf{Q}} &= \min_{\mathbf{Q}}\ \lambda_2\|\Theta\odot\mathbf{Q}\|_1+\left\langle\hat{\mathbf{Y}}_3^t,\mathbf{Q}-\mathbf{Z}^{t+1}\right\rangle\\ &\quad+\frac{\mu^t}{2}\|\mathbf{Q}-\mathbf{Z}^{t+1}\|_F^2\\ &= \lambda_2\|\Theta\odot\mathbf{Q}\Big\|_1+\frac{\mu^t}{2}\|\mathbf{Q}-\left(\mathbf{Z}^{t+1}-\frac{\hat{\mathbf{Y}}_3^t}{\mu^t}\right)\Big\|_F^2. \end{aligned} \quad (20)$$

Equation (20) can be updated with an elementwise strategy, and the problem can be equivalently decomposed into $n\times N$ subproblems. Then, the optimal solution to (20) can be obtained as follows:

$$\begin{aligned}L_{\mathbf{Q}} &= \min_{\mathbf{Q}}\ \lambda_2\Theta_{ij}|\mathbf{Q}_{ij}|+\frac{\mu^t}{2}\left(\mathbf{Q}_{ij}-\left(\mathbf{Z}^{t+1}-\frac{\hat{\mathbf{Y}}_3^t}{\mu^t}\right)_{ij}\right)^2\\ &= \hat{S}_{\lambda_2\Theta_{ij}/\mu^t}\left(\mathbf{Z}^{t+1}-\frac{\hat{\mathbf{Y}}_3^t}{\mu^t}\right)_{ij}. \end{aligned} \quad (21)$$

4) E is updated by converting the objective function [see (15)] into the following form [see (22)]:

$$\begin{aligned}L_{\mathbf{E}} &= \min_{\mathbf{E}}\ \lambda_4\|\mathbf{E}\|_{2,1}+\langle\hat{\mathbf{Y}}_1,\mathbf{X}-\mathbf{X}_{\mathrm{tr}}\mathbf{Z}^{t+1}-\mathbf{E}\rangle\\ &\quad+\frac{\mu^t}{2}\|\mathbf{X}-\mathbf{X}_{\mathrm{tr}}\mathbf{Z}^{t+1}-\mathbf{E}\|_F^2\\ &= \lambda_4\|\mathbf{E}\|_{2,1}+\frac{\mu^t}{2}\left\|\mathbf{E}-\left(\mathbf{X}-\mathbf{X}_{\mathrm{tr}}\ \mathbf{Z}^{t+1}+\frac{\hat{\mathbf{Y}}_1}{\mu^t}\right)\right\|_F^2. \end{aligned} \quad (22)$$

which can be solved by Lemma 1.

Lemma 1 [19]: Let $\boldsymbol{W}$ be a given matrix. The optimization problem with respect to $\boldsymbol{G}$ can be formulated as follows:

$$\min_{\boldsymbol{G}}\ \alpha\|\boldsymbol{G}\|_{2,1}+\frac{1}{2}\|\boldsymbol{W}-\boldsymbol{G}\|_{\mathrm{F}}^2. \quad (23)$$

Then, the optimal solution $\boldsymbol{G}^*$ can be written in the following form:

$$\boldsymbol{G}_i^* = \begin{cases}\dfrac{\|\boldsymbol{G}_i\|_2-\alpha}{\|\boldsymbol{G}_i\|_2}\boldsymbol{W}_i, & \text{if}\|\boldsymbol{W}_i\|_2>0\\ 0, & \text{otherwise}\end{cases} \quad (24)$$

where $\boldsymbol{G}_i$ and $\boldsymbol{W}_i$ are the ith columns of matrices G and $\boldsymbol{W}$, respectively.

The Lagrangian multipliers $\hat{\boldsymbol{Y}}_1$, $\hat{\boldsymbol{Y}}_2$, and $\hat{\boldsymbol{Y}}_3$, and penalty parameter μ also need to be updated via the ADMM algorithm after $\boldsymbol{Z}$, $\boldsymbol{P}$, $\boldsymbol{Q}$, and E are updated. The detailed procedure for solving the SSBDFCP model (12) is described in Algorithm 1.

Algorithm 1 Solve the SSBDFCP Model via the ADMM

Input: The hyperspectral image matrix $\boldsymbol{X} = [\boldsymbol{X}_{\mathrm{tr}}, \boldsymbol{X}_{\mathrm{tt}}]$; Parameters λ_1, λ_2, λ_3, λ_4; Fuzzy class probability measure matrix Θ.

Initialization: $\boldsymbol{Z}=\mathbf{0}$, $\boldsymbol{P}=\mathbf{0}$, $\boldsymbol{Q}=\mathbf{0}$, $\boldsymbol{E}=\mathbf{0}$, $\hat{\boldsymbol{Y}}_1=\mathbf{0}$, $\hat{\mathbf{Y}}_2=\mathbf{0}$, $\hat{\mathbf{Y}}_3=\mathbf{0}$, $\lambda_1,\lambda_2,\lambda_3,\lambda_4>0$, $\mu_{\max}=1e8$, $\rho=1.15$, $tol=1e-6$.

While not converged do

1) Update $\boldsymbol{Z}$ and fix the others by solving (18);
2) Update $\boldsymbol{P}$ and fix the others by solving (19);
3) Update $\boldsymbol{Q}$ and fix the others by solving (21);
4) Update $\boldsymbol{E}$ and fix the others by solving (22);
5) Update Lagrange multipliers $\hat{\mathbf{Y}}_1$, $\hat{\mathbf{Y}}_2$ and $\hat{\mathbf{Y}}_3$:

$$\begin{cases}\hat{\mathbf{Y}}_1^{t+1}=\hat{\mathbf{Y}}_1^{t+1}+\mu^t(\mathbf{X}-\mathbf{X}_{\mathrm{tr}}\mathbf{Z}^{t+1}-\mathbf{E}^{t+1})\\ \hat{\mathbf{Y}}_2^{t+1}=\hat{\mathbf{Y}}_2^{t+1}+\mu^t(\mathbf{P}^{t+1}-\mathbf{Z}^{t+1})\\ \hat{\mathbf{Y}}_3^{t+1}=\hat{\mathbf{Y}}_3^{t+1}+\mu^t(\mathbf{Q}^{t+1}-\mathbf{Z}^{t+1}).\end{cases}$$

7) Update μ:
$\mu^{t+1}=\min(\mu_{\max},\rho\mu^t)$.
8) Check the convergence conditions
$\max\left(\begin{matrix}\|\mathbf{X}-\mathbf{X}_{\mathrm{tr}}\mathbf{Z}^{t+1}-\mathbf{E}^{t+1}\|_\infty,\\ \|\mathbf{P}^{t+1}-\mathbf{Z}^{t+1}\|_\infty,\|\mathbf{Q}^{t+1}-\mathbf{Z}^{t+1}\|_\infty\end{matrix}\right)\le tol$.
9) Update k: $k\leftarrow k+1$.

End while

Output: $\boldsymbol{Z}$, $\boldsymbol{P}$, $\boldsymbol{Q}$, $\boldsymbol{E}$.

The SSBDFCP model (12) can be optimized via Algorithm 1, and the spatial–spectral block-diagonal discriminative representation of an HSI $\mathbf{Z}=[\mathbf{Z_{tr}},\mathbf{Z_{tt}}]$ can be obtained. Given the label indicator binary matrix $\mathbf{Y}\in\mathbb{R}^{l\times C}$ and the block-diagonal representation of labeled samples $\mathbf{Z_{tr}}\in\mathbb{R}^{l\times l}$, then the final recognition method [25] is obtained by a simple linear classifier [24], which can be described as follows.

The discriminant data representation $\boldsymbol{Z}=[\mathbf{Z}_{\mathrm{tr}},\mathbf{Z}_{\mathrm{tt}}]$ can be obtained by optimizing the problem (12) via the ADMM. The linear classifier $\hat{\boldsymbol{W}}$ is learned from the training data representation and its corresponding label matrix $\boldsymbol{Y}\in\mathbb{R}^{l\times C}$ of the training samples. The learned linear classifier can be defined as follows:

$$\hat{\boldsymbol{W}}^* = \arg\min_{\hat{\boldsymbol{W}}}\|\boldsymbol{Y}^T-\hat{\boldsymbol{W}}\mathbf{Z}_{\mathrm{tr}}\|_F^2+\gamma\|\hat{\boldsymbol{W}}\|_F^2 \quad (25)$$

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where γ is a positive regularization parameter. The closed-form solution can be easily obtained

$$\hat{\boldsymbol{W}}^* = \boldsymbol{Y}^T \boldsymbol{Z}_{\mathrm{tr}}^T (\boldsymbol{Z}_{\mathrm{tr}} \boldsymbol{Z}_{\mathrm{tr}}^T + \gamma \mathbf{I})^{-1}. \tag{26}$$

The recognition of test samples is determined as follows:

$$\text{label } (\boldsymbol{X}_{\mathrm{tt}}) = \arg\max(\hat{\boldsymbol{W}}^* \boldsymbol{Z}_{\mathrm{tt}}). \tag{27}$$

B. Discussion

The SSBDFCP model simultaneously takes advantages of sparse, low-rank, and FCP structures, as well as weakly feedback information, to learn the block-diagonal representation of an HSI. The relationships between the SSBDFCP model and some related previous works can be established.

1) *Connection to the PCSSR Model:* The PCSSR model utilizes partial labeled information and abundant unlabeled samples to learn probabilistic class structure information via SR, which reflects the probabilistic relationship between each sample and each class. Then, the probabilistic class structure information is incorporated into the SR model to yield discriminative representation coefficients for image classification. The PCSSR model is formulated as follows:

$$\min_{\boldsymbol{Z}} \frac{1}{2} \|\boldsymbol{X} - \boldsymbol{X}_{\mathrm{tr}} \boldsymbol{Z}\|_{\mathrm{F}}^2 + \|\Theta' \odot \boldsymbol{Z}\|_1 \tag{28}$$

where $\Theta' = (\lambda_1 1^T + \lambda_2 \hat{\Theta})$. Suppose that the low-rank and block-diagonal constraints on the SSBDFCP model are eliminated and that the integrated HSI spatial information is not considered. In such a case, the SSBDFCP model can be reduced to the following form:

$$\min_{\mathbf{Z},\mathbf{E}} \lambda_2 \|\Theta \odot \mathbf{Z}\|_1 + \lambda_4 \|\mathbf{X} - \mathbf{X}_{\mathrm{tr}} \mathbf{Z}\|_{2,1}. \tag{29}$$

Equation (29) is approximately the same as the PCSSR model (28) in practice without the two main constraints of the SSBDFCP model. The PCSSR is a special case of the SSBDFCP model. Therefore, the PCSSR cannot present the low-rank and block-diagonal characteristics of representation coefficients and performs poorly for intraclass similar samples. Moreover, the discriminative ability of the model is poor when unfavorable features and noise are present based on the self-expressiveness property of the method.

2) *Connection to the NNLRS Model:* The NNLRS model jointly integrates sparse and low-rank models into a unified framework to create an informative sparse and low-rank graph that can capture the global and local structures of original samples. The NNLRS model can be formulated as follows:

$$\begin{aligned} &\min_{\mathbf{Z},\mathbf{E}} \|\mathbf{Z}\|_* + \beta \|\mathbf{Z}\|_1 + \lambda \|\mathbf{E}\|_{2,1} \\ &\text{s.t. } \mathbf{X} = \mathbf{X}_{\mathrm{tr}} \mathbf{Z} + \mathbf{E}. \end{aligned} \tag{30}$$

The SSBDFCP model sets $\lambda_3 \to 0$, and the measurements of the FCP and spatial information are not considered. Furthermore, the block-diagonal representation is eliminated with $\lambda_1 \to 0$. Therefore, the reduced model considering the aforementioned operations can only capture the global and local structure information and is reformulated as follows:

$$\begin{aligned} &\min_{\mathbf{Z},\mathbf{E}} \|\mathbf{Z}\|_* + \lambda_2 \|\mathbf{Z}\|_1 + \lambda_4 \|\mathbf{E}\|_{2,1} \\ &\text{s.t. } \mathbf{X} = \mathbf{X}_{\mathrm{tr}} \mathbf{Z} + \mathbf{E}. \end{aligned} \tag{31}$$

The NNLRS model is also a special case of the SSBDFCP model that is not robust to capturing the spatial information of an HSI and suppressing the off-block-diagonal components, which should be as small as possible. Moreover, the SSBDFCP model is more likely to benefit from the true neighbors based on its self-expressiveness property. This approach can remove noise, and weakly supervised information is used for feedback in each iteration.

C. Convergence Analysis

Solving for $\boldsymbol{Z}$, $\boldsymbol{P}$, $\boldsymbol{Q}$, and $\boldsymbol{E}$ simultaneously with (12) is very difficult but can be accomplished with the ADMM algorithm, as shown in Algorithm 1. According to [24], the optimal solution method for $\boldsymbol{Z}$, $\boldsymbol{P}$, $\boldsymbol{Q}$, and $\boldsymbol{E}$ involves a closed proper convex function. With the auxiliary variable added, the optimization of $\boldsymbol{Z}$ has a closed form. The other variables $\boldsymbol{P}$ and $\boldsymbol{Q}$ are optimized with the singular-value thresholding. The variable $\boldsymbol{E}$ can be obtained with a closed-form process [19]. Moreover, the optimization of the four variables $\boldsymbol{Z}$, $\boldsymbol{P}$, $\boldsymbol{Q}$, and $\boldsymbol{E}$ is equivalent to solving a two-block ADMM problems with theoretically guaranteed global convergence [40]. Subsequently, the convergence of the SSBDFCP solved with the ADMM is demonstrated in Algorithm 1.

D. Computational Complexity Analysis

Clearly, the computational complexity of the SSBDFCP model is mainly associated with Algorithm 1, and the major computational complexity of Algorithm 1 is presented. Specifically, the computational cost of Algorithm 1 is high in steps 1–4. The computational complexity of calculating $\boldsymbol{Z}$ is $\mathrm{O}(dl^2 + Nl^2)$ due to the matrix inverse operation, where d is the dimensionality of the samples. The computational complexity of calculating $\boldsymbol{P}$ is approximately $\mathrm{O}(Nl^2)$, $(N > l)$ due to the singular value decomposition (SVD) of the matrix. The SVD is time-consuming when the number of labeled samples l and the total number of samples N are very large. In addition, computing $\boldsymbol{Q}$ is associated with a computational complexity of $\mathrm{O}(Nl)$ because of solving the l_1 problem. Calculating $\boldsymbol{E}$ in step 4 yields a computational complexity of $\mathrm{O}(dN)$. In summary, the total computational complexity of the SSBDFCP model is approximately $\mathrm{O}(t(dl^2 + 2Nl^2 + Nl + dN))$, where t is the number of iterations. In comparison, the computational complexity of the PCSSR model is approximately $\mathrm{O}(tN^3)$ as this approach mainly depends on performing inverse operations for all samples. The computational complexity of SCSSR is higher than that of PCSSR because the spatial Laplacian matrix is used in the SCSSR model. Moreover, the NNLRS model must implement the SVD of the feature matrix and simultaneously solve the simple soft-thresholding problem. Therefore, the computational complexity of the proposed SSBDFCP model is similar to the complexity of other sparse, LRR models but lower than that of the PCSSR and SCSSR methods.

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E. Potential Capability of SSBDFCP for an Out-of-Sample of HSI

As we know, given unobserved samples outside the training and testing samples, reimplementing the whole model for classification of novel HSI samples would be time-consuming. Notably, SSBDFCP can be extended to address the out-of-sample problem in HSI classification and can manage new HSI samples.

Suppose that the optimal block-diagonal representation $\boldsymbol{Z} \in \mathbb{R}^{l\times N}$ is obtained via SSBDFCP from the available HSI samples $\boldsymbol{X} = [\boldsymbol{X}_{\rm tr}, \boldsymbol{X}_{\rm tt}] \in \mathbb{R}^{d\times N}$ using the proposed model (12). Given an unobserved sample $\boldsymbol{b} \in \mathbb{R}^{d\times 1}$ outside the training and testing samples in the original observed space, we aim to learn the discriminative representation $\boldsymbol{z}^b$ for $\boldsymbol{b}$ over $\boldsymbol{X}_{\rm tr}$ when fixing the previously learned representation $\boldsymbol{Z}$. Then, two cases with and without the new sample $\boldsymbol{b}$ in the spatially local window $s \times s$ of training samples can be considered.

1) Assume that a new sample $\boldsymbol{b}$ is far away from the training and testing samples and its spatial neighbors are without the spatially local window $s \times s$ of the training samples. Therefore, by adding the new sample $\boldsymbol{b}$ in model (12) and fixing the other variables, the objective function of the augmented SSBDFCP is formulated as

$$\begin{aligned}\min_{\mathbf{Z},\mathbf{E},\mathbf{W}}\ & \|[\mathbf{Z},\mathbf{z}^b]\|_* + \lambda_1\|\breve{\mathbf{A}}\odot[\mathbf{Z},\mathbf{z}^b]\|_F^2 \\ &+ \lambda_2\|\tilde{\Theta}\odot[\mathbf{Z},\mathbf{z}^b]\|_1 \\ &+ \lambda_3\|[\mathbf{Z},\mathbf{z}^b]-[\mathbf{S},\mathbf{0}_l]\|_F^2 + \lambda_4\|[\mathbf{E},e]\|_{2,1} \\ \text{s.t. }& [\mathbf{X},\mathbf{b}] = \mathbf{X}_{\rm tr}[\mathbf{Z},\mathbf{z}^b] + [\mathbf{E},e] \end{aligned} \tag{32}$$

where $\breve{\mathbf{A}} = [\tilde{\mathbf{A}}, 1_l 1_{N+1-l}^T]$ and e is the noise. The initialization phase of the FCP $\tilde{\Theta} \in \mathbb{R}^{l\times(N+1)}$ can be obtained by the fuzzy K-nearest neighbor (FKNN) algorithm [34] in the same manner as the SSBDFCP model over the training samples $\mathbf{X}_{\rm tr}$ and all samples $[\mathbf{X},\mathbf{b}]$. The feedback of FCP plays an important role after the second iteration based on the model (9). In general, for the learned representation $\mathbf{Z} \in \mathbb{R}^{l\times N}(l < N)$, the singular values of matrix $\mathbf{Z}$ coincide with $[\mathbf{Z},\mathbf{z}^b]$ and $\|[\mathbf{Z},\mathbf{z}^b]\|_* = \|\mathbf{Z}\|_*$, whose more detailed proof can found in [24]. Therefore, the objective function (32) with respect to $\mathbf{z}^b$ and e can be rewritten as

$$\begin{aligned}&\min_{\mathbf{z}^b,e}(\lambda_1+\lambda_3)\|\mathbf{z}^b\|_2^2 + \lambda_2\|\tilde{\Theta}_b\odot\mathbf{z}^b\|_1 + \lambda_4\|e\|_2^2 \\ &\text{s.t. } \mathbf{b} = \mathbf{X}_{\rm tr}\mathbf{z}^b + e \end{aligned} \tag{33}$$

where $\tilde{\Theta}_b \in \mathbb{R}^{l\times 1}$. To clarify problem (33) clear, it is also reformulated as

$$\min_{\mathbf{z}^b,e}\frac{1}{2}\|\mathbf{b}-\mathbf{X}_{\rm tr}\mathbf{z}^b\|_2^2 + \frac{\beta_1}{2}\|\mathbf{z}^b\|_2^2 + \beta_2\|\tilde{\Theta}_b\odot\mathbf{z}^b\|_1 \tag{34}$$

where $\beta_1 = (\lambda_1+\lambda_3)/\lambda_4$ and $\beta_2 = \lambda_2/2\lambda_4$. For convenience, denote $g(\mathbf{z}^b) = (1/2)\|\mathbf{b}-\mathbf{X}_{\rm tr}\mathbf{z}^b\|_2^2 + (\beta_1/2)\|\mathbf{z}^b\|_2^2$. According to [41], problem (34) can be approximately transformed to the following optimization problem:

$$\begin{aligned}\mathbf{z}_{k+1}^b &= \arg\min_{\mathbf{z}^b}\beta_2\|\tilde{\Theta}_b\odot\mathbf{z}^b\|_1 \\ &\quad + \langle\nabla_{\mathbf{z}^b}g(\mathbf{z}^b), \mathbf{z}^b-\mathbf{z}_k^b\rangle + \frac{\eta}{2}\|\mathbf{z}^b-\mathbf{z}_k^b\|_2^2 \\ &= \arg\min_{\mathbf{z}^b}\beta_2\|\tilde{\Theta}_b\odot\mathbf{z}^b\|_1 \\ &\quad + \frac{\eta}{2}\|\mathbf{z}^b-\mathbf{z}_k^b+\nabla_{\mathbf{z}^b}g(\mathbf{z}^b)/\eta\|_2^2 \end{aligned} \tag{35}$$

where $\mathbf{z}_k^b$ is the k iteration of $\mathbf{z}^b$ and η is a balance parameter. Based on model (21), the optimal solution of $\mathbf{z}^b$ can be obtained by using $\mathbf{z}_{k+1}^b = \hat{S}_{\beta_2\tilde{\Theta}_b/\eta}(\mathbf{z}_k^b - \nabla_{\mathbf{z}^b}g(\mathbf{z}_k^b)/\eta)$.

2) Assume that a new sample $\boldsymbol{b}$ is within the spatially local window $s\times s$ of the training samples. Then, the objective function of SSBDFCP is formulated as

$$\begin{aligned}\min_{\mathbf{Z},\mathbf{E},\mathbf{W}}\ & \|[\mathbf{Z},\mathbf{z}^b]\|_* + \lambda_1\|\breve{\mathbf{A}}\odot[\mathbf{Z},\mathbf{z}^b]\|_F^2 \\ &+ \lambda_2\|\tilde{\Theta}\odot[\mathbf{Z},\mathbf{z}^b]\|_1 \\ &+ \lambda_3\|[\mathbf{Z},\mathbf{z}^b]-[\mathbf{S},\mathbf{s}^b]\|_F^2 + \lambda_4\|[\mathbf{E},e]\|_{2,1} \\ \text{s.t. }& [\mathbf{X},\mathbf{b}] = \mathbf{X}_{\rm tr}[\mathbf{Z},\mathbf{z}^b] + [\mathbf{E},e] \end{aligned} \tag{36}$$

where $\mathbf{s}^b \in \mathbb{R}^{l\times 1}$ is the spatial information expression within local window. Therefore, the objective function (36) with respect to $\mathbf{z}^b$ and e can be rewritten as

$$\begin{aligned}&\min_{\mathbf{z}^b,e}(\lambda_1+\lambda_3)\|\mathbf{z}^b-\mathbf{s}^b/(\lambda_1+\lambda_3)\|_2^2 \\ &\quad + \lambda_2\|\tilde{\Theta}_b\odot\mathbf{z}^b\|_1 + \lambda_4\|e\|_2^2 \\ &\text{s.t. } \mathbf{b} = \mathbf{X}_{\rm tr}\mathbf{z}^b + e. \end{aligned} \tag{37}$$

Similar to problem (33), the optimal solution of $\mathbf{z}^b$ can also be obtained.

After obtaining the optimal solution $\mathbf{z}^b$, a new HSI sample $\boldsymbol{b}$ can be identified by using label $(\boldsymbol{b}) = \arg\max(\hat{\boldsymbol{W}}^*\boldsymbol{z}^b)$. Based on the earlier analysis, a new HSI sample $\boldsymbol{b}$ can obtain a better recognition result. Therefore, based on the proposed SSBDFCP model, the new HSI sample outside the training and testing samples can be sufficiently addressed.

V. Experimental Results and Discussion

In this section, we quantitatively and qualitatively evaluate the proposed SSBDFCP model based on two real benchmark HSIs for classification. First, the experimental data are briefly introduced. Then, the related graph approaches, including the BDLRR, PCSSR, SCSSR, LLR, GK, JSR [42], and SVMCK [43] methods as well as the EMP [3], EMAP [4], and edge-preserving filtering (EPF) [44] methods, are compared with the SSBDFCP model.

A. Data Set Description

The experiments were performed based on two typical data sets: the Indian Pines and the PaviaU data sets [45].

1) *Indian Pines:* This data set was obtained over Northwestern Indiana with an AVIRIS sensor in 1992. The sensor consists of 145 × 145 pixels and 220 spectral bands within the range of 375 − 2500 nm. The images in the data set contain 16 land-cover types.

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Fig. 3. Indian Pines data. (a) False-color image and (b) ground-truth image (bands 20, 69, and 178 are red, green and blue, respectively).

TABLE I
NUMBERS OF TRAINING AND TESTING SAMPLES ON THE INDIAN PINES AND PAVIAU DATA SETS

Indian Pines				PaviaU			
Class	Name	train	test	Class	Name	Train	Test
1	Alfalfa	9	37	1	Asphalt	596	2385
2	Corn-notill	285	1143	2	Meadows	1351	5406
3	Corn-mintill	166	664	3	Gravel	403	1614
4	Corn	47	190	4	Trees	239	960
5	Grass-pasture	96	387	5	Metal sheet	269	1076
6	Grass-trees	146	584	6	Bare Soil	1001	4005
7	Grass-pasture-mowed	5	23	7	Bitumen	266	1064
8	Hay-windrowed	95	383	8	Bricks	571	2286
9	Oats	4	16	9	Shadows	188	753
10	Soybean-notill	194	778		Total	4884	19549
11	Soybean-mintill	491	1964				
12	Soybean-clean	118	475				
13	Wheat	41	164				
14	Woods	253	1012				
15	Buildings-Grass-Trees-Drives	77	309				
16	Stone-Steel-Towers	18	75				
	Total	2045	8204				

In the experiment, the SSBDFCP model and the other compared methods were tested using the entire data set. The false color and corresponding ground-truth images of Indian Pines are shown in Fig. 3.

2) *PaviaU:* This data set was acquired at Pavia University by the ROSIS sensor during a flight campaign over Pavia University, Northern Italy, in 2002. The images are 610×340 pixels with 103 spectral bands. Moreover, the geometric resolution of the sensor was 1.3 m, and nine land-cover types are included in the data set. In the experiment, the SSBDFCP model and the other methods were tested based on a subset of the PaviaU data set with 457×231 pixels and 103 spectral bands. The false color and corresponding ground-truth images of PaviaU are shown in Fig. 4.

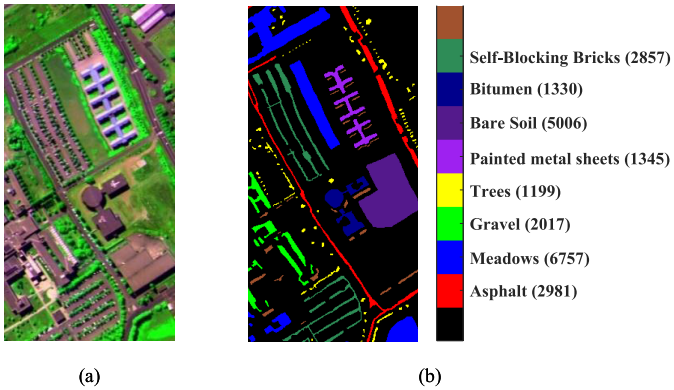


Fig. 4. Subset of the PaviaU data. (a) False-color image and (b) ground-truth image (bands 52, 99, and 4 are red, green and blue, respectively).

B. Experimental Setup

To explore the performance of the SSBDFCP method, each data set was randomly divided into the labeled and unlabeled samples. Then, we randomly selected 20% of the samples in each class from each data set as labeled samples, and the remaining samples were treated as unlabeled samples unless otherwise noted. The detailed numbers of training and testing samples used are given in Table I. Moreover, each method was run five times with randomly selected samples to test the performance of each data set. Note that the number of nearest neighbors was set to $K = 5$ [34]. Finally, we used the overall accuracy (OA), kappa coefficient (KC), and class accuracy (CA) to evaluate the results of each method.

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TABLE II
HSI CLASSIFICATION RESULTS (%) FOR EACH CLASS IN THE INDIAN PINES DATA SET

Class	GK	LLR	SVMCK	JSR	EMP	EMAP	EPF	PCSSR	SCSSR	BDLRR	SSBDCP	SSBDFCP
Alfalfa	43.24	18.92	**100**	96.55	75.68	51.35	**100**	47.83	19.57	32.43	**100**	97.30
Corn-no till	60.28	59.67	98.42	88.95	87.58	91.78	97.48	58.26	50.35	75.85	98.60	**99.04**
Corn-min till	51.81	50.90	**99.55**	91.51	91.57	89.46	98.82	42.89	29.04	51.51	99.10	98.04
Corn	35.79	23.16	**99.47**	96.69	79.47	88.42	93.72	40.93	20.25	27.37	**99.47**	97.37
Grass-pasture	84.50	82.95	97.21	97.08	96.12	96.64	97.95	78.88	64.80	88.89	**98.97**	97.42
Grass-trees	95.72	97.26	98.81	96.80	98.46	98.80	95.73	92.33	95.34	98.80	**99.66**	98.29
Grass-pasture-mowed	86.96	82.61	90.48	**100**	65.22	65.22	**100**	53.57	17.86	56.52	78.26	**100**
Hay-windrowed	98.43	99.22	99.22	97.93	**100**	99.74	92.96	99.16	92.89	99.48	**100**	99.74
Oats	31.25	12.50	**100**	**100**	50.00	75.00	0	50.00	20.00	25.00	87.50	**100**
Soybean-no till	71.59	73.01	97.29	90.75	86.38	80.98	95.95	52.37	39.92	63.88	99.61	**100**
Soybean-min till	78.00	83.15	98.68	84.39	96.44	92.52	92.98	88.31	97.11	81.87	98.52	**99.34**
Soybean-clean	43.16	46.11	98.11	91.45	87.58	78.11	93.91	47.39	25.80	61.47	98.74	**100**
Wheat	92.07	95.12	**100**	98.17	98.78	96.95	**100**	93.66	93.17	98.17	**100**	**100**
Woods	94.47	96.05	**99.90**	97.56	97.92	98.52	97.78	97.55	99.68	97.73	**99.90**	**99.90**
Buildings-Grass-Trees-Drives	39.16	37.86	99.67	99.31	94.82	90.61	98.29	49.74	21.50	47.23	98.71	**99.68**
Stone-Steel-Towers	94.67	85.33	**98.67**	80.68	94.67	93.33	88.89	76.34	39.78	86.67	92.00	97.33
OA	73.11	74.23	98.73	91.29	93.13	91.59	95.70	73.26	68.07	77.40	98.96	**99.17**
KC	0.69	0.70	0.98	0.90	0.92	0.90	0.95	0.69	0.62	0.74	**0.99**	**0.99**

TABLE III
HSI CLASSIFICATION RESULTS (%) FOR EACH CLASS IN THE PAVIAU DATA SET

Class	GK	LLR	SVMCK	JSR	EMP	EMAP	EPF	PCSSR	SCSSR	BDLRR	SSBDCP	SSBDFCP
Asphalt	76.77	77.11	98.84	99.79	98.74	98.78	96.48	22.34	22.44	83.86	**99.96**	99.83
Meadows	94.51	95.45	**99.74**	99.76	99.19	98.46	99.00	94.67	92.64	94.88	99.59	99.67
Gravel	65.43	66.42	98.82	96.87	98.82	99.44	99.12	20.77	20.48	64.56	99.81	**99.94**
Trees	93.33	91.77	99.05	99.26	98.33	93.54	**100**	20.85	20.85	92.40	99.17	98.33
Painted metal sheets	99.81	99.44	**100**	98.35	99.91	99.63	**100**	20.89	21.12	**100**	**100**	99.81
Bare soil	83.60	84.42	99.80	99.92	98.50	99.90	98.14	64.32	65.58	96.53	**100**	**100**
Bitumen	84.96	85.15	**100**	**100**	98.78	99.91	**100**	20.38	20.23	86.56	99.15	99.53
Self-blocking bricks	84.60	85.52	94.55	99.43	98.64	91.03	94.15	20.48	20.37	81.15	98.86	**99.61**
Shadows	99.07	98.94	**100**	**100**	99.87	99.87	98.30	22.53	25.61	96.68	97.74	97.21
OA	86.44	87.01	98.98	99.43	98.90	97.96	98.04	50.35	50.14	89.54	99.56	**99.61**
KC	0.84	0.84	0.98	**0.99**	0.98	0.98	0.98	0.35	0.35	0.87	**0.99**	**0.99**

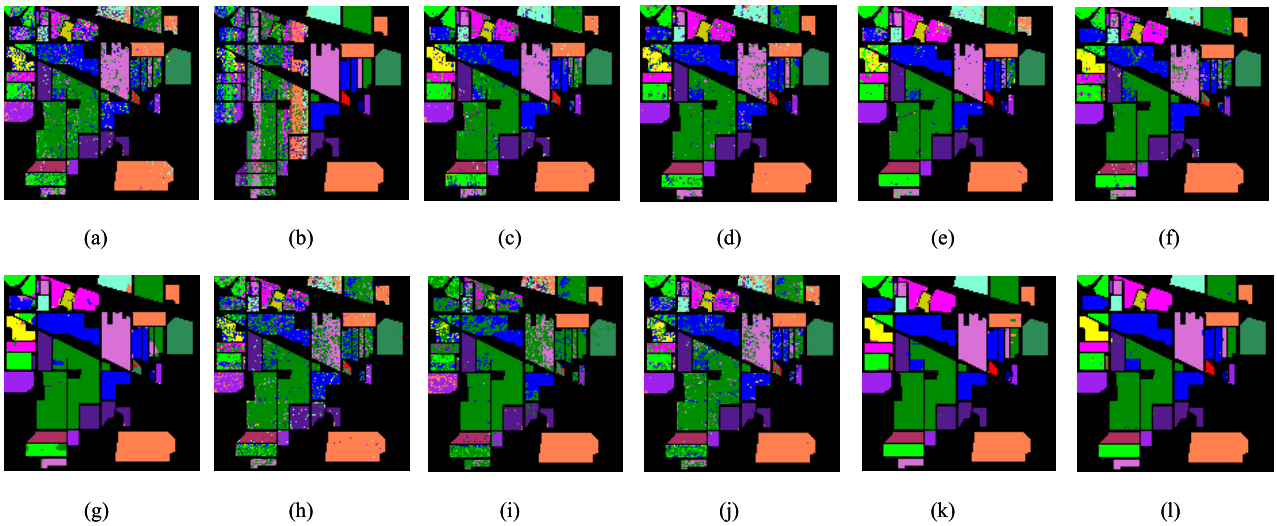
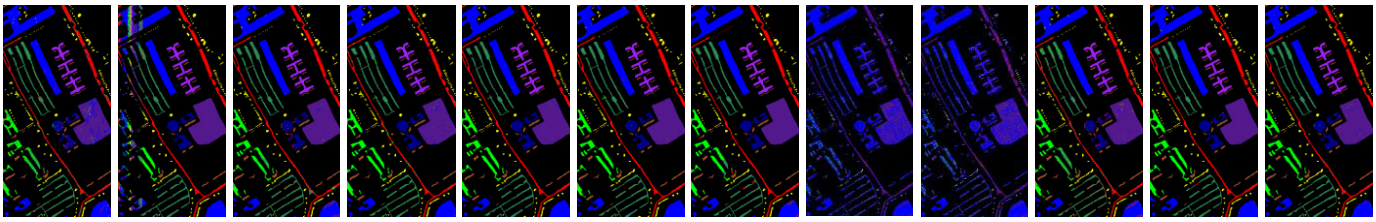


Fig. 5. Classification maps of different graph-based methods for the Indian Pines data set. (a) GK. (b) LLR. (c) SVMCK. (d) JSR. (e) EMP. (f) EMAP. (g) EPF. (h) PCSSR. (i) SCSSR. (j) BDLRR. (k) SSBDCP. (l) SSBDFCP.

C. Experimental Results

The quantitative classification results of the seven graph-based methods for the Indian Pines and PaviaU data sets are listed in Tables II and III and shown in Figs. 5 and 6. The classification performance of different graph methods run five times is presented in Tables IV and V.

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 6. Classification results for different graph-based methods based on the PaviaU data set. (a) GK. (b) LLR. (c) SVMCK. (d) JSR. (e) EMP. (f) EMAP. (g) EPF. (h) PCSSR. (i) SCSSR. (j) BDLRR. (k) SSBDCP. (l) SSBDFCP.

TABLE IV
CLASSIFICATION RESULTS WITH DIFFERENT RUN TIMES FOR INDIAN PINES [OA(STD) (%), KC(STD)

Methods	GK	LLR	SVMCK	JSR	EMP	EMAP	EPF	PCSSR	SCSSR	BDLRR	SSBDCP	SSBDFCP
OA	71.15(1.71)	73.98(0.25)	98.63(0.22)	90.74(0.94)	92.84(0.25)	91.33(0.63)	95.60(0.77)	72.77(0.50)	67.77(0.21)	76.22(0.80)	98.65(0.30)	**99.09(0.14)**
KC	0.67(0.02)	0.70(0.00)	0.98(0.25)	0.89(1.13)	0.92(0.58)	0.90(0.72)	0.95(0.89)	0.68(0.01)	0.61(0.00)	0.72(0.00)	0.98(0.33)	**0.99(0.16)**

TABLE V
CLASSIFICATION RESULTS WITH DIFFERENT RUN TIMES FOR PAVIAU [OA(%) (STD), KC(STD)]

Methods	GK	LLR	SVMCK	JSR	EMP	EMAP	EPF	PCSSR	SCSSR	BDLRR	SSBDCP	SSBDFCP
OA	85.30(1.03)	86.81(0.17)	98.91(0.33)	99.34(0.05)	98.85(0.26)	97.60(0.37)	97.84(0.19)	50.57(0.45)	50.47(0.71)	88.82(0.48)	99.42(0.13)	**99.53(0.17)**
KC	0.82(0.01)	0.84(0.00)	0.98(0.40)	**0.99(0.05)**	0.98(0.31)	0.97(0.44)	0.97(0.22)	0.36(0.01)	0.36(0.01)	0.87(0.01)	**0.99(0.14)**	**0.99(0.21)**

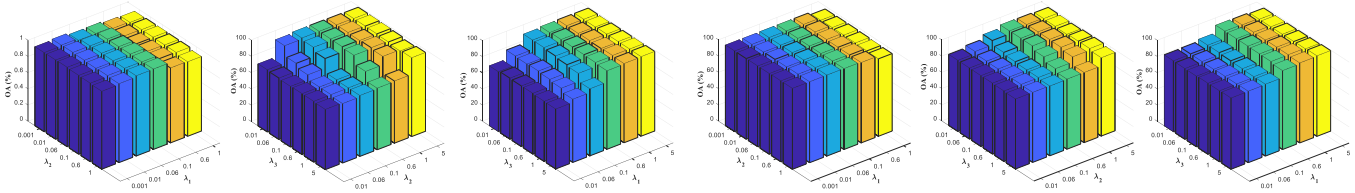
Table II shows that the OA and KC for the SSBDFCP method are higher than those for the other methods, and the proposed method achieves an improvement of approximately 3% compared with the EPF method. Moreover, the SSBDFCP and SSBDCP methods generally perform better than the other methods due to the guidance provided by labeled information in the block structure graph models, which is critical for HSI classification. SSBDFCP and SSBDCP can learn based on true neighbors by removing much of the noise around pixels via robust feature learning. In particular, compared with SCSSR, SVMCK, and JSR, the rich spatial information can be better utilized for HSI classification in the SSBDFCP model, which is derived from the integration of spatial information into the block structure. Based on the OA and KC values, the SSBDFCP method performs better than the SSBDCP method, indicating that SSBDFCP can capture the useful latent subspace structural relationships and propagate the pure labeled information to unlabeled samples. In other words, the FCP efficiently considers the probability of a sample belonging to each class and propagates the labeled information to unlabeled samples rather than using the class probability obtained by SR. Thus, the proposed approach can better reveal the similarity of the subspace structure by using the weakly supervised information as feedback, that is, a correctly classified sample has a membership value in the corresponding class close to one.

Compared with the BDLRR method, the SSBDFCP method achieves superior results, as illustrated by both Fig. 5(j) and 5(l) and the OA and KC values. The main reason for this result is that SSBDFCP can enhance the discriminative recognition ability of the intraclass representation because of the guidance provided by labeled information and robust learning; therefore, this approach differs from methods using the original unsupervised Euclidean distance constraint. As shown in Fig. 5, although the classification precision of each class in the Indian Pines data set is not always high based on the SSBDFCP method, most land-cover types are correctly classified, except for Oats and Alfalfa. Notably, spectral information for Oats is similar to that for Grass-trees, and Oats are commonly misclassified as Grass-trees. Moreover, the visual classification results based on the SSBDFCP method are closer to the ground-truth data than the results of other methods.

Table III shows that the overall classification precision for the PaviaU data set is highest with the proposed SSBDFCP method compared with that of the other methods, with the maximum OA and KC values of 99.61% and 0.99, respectively. Compared with the SVMCK, JSR, EMP, EMAP, and EPF methods, the SSBDFCP method includes discriminative recognition in HSI classification and obtains a better intraclass presentation. Based on the experimental results, the SSBDFCP method can propagate fully labeled information to unlabeled samples under the guidance of the latent subspace structure representation with the FCP, which provides an efficiency constraint on the block-diagonal structure, in contrast to the SSBDCP method. Moreover, the FCP feedback can be used to effectively build the relationship between the block structure and fuzzy memberships and facilitate the similarity expression among block structure coefficient vectors.

As shown in Fig. 6, the classification precision of every class in PaviaU is high, although optimal recognition cannot be guaranteed. Fortunately, the classification precision of each class is greater than 95% based on the proposed method, which provides a classification map that is very similar to the ground-truth data.

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(a) (b) (c) (d) (e) (f)

Fig. 7. OAs of different tradeoff parameters for (a)–(c) Indian Pines and (d)–(f) PaviaU data sets.

TABLE VI
OPTIMAL PARAMETERS OF DIFFERENT METHODS FOR THE INDIAN PINES AND PAVIAU DATA SETS

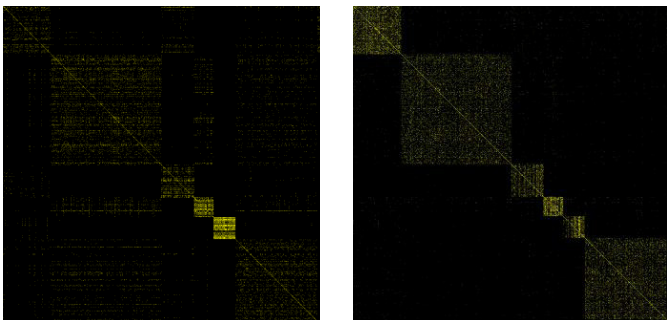
Methods	Indian Pines	PaviaU
GK	$k=5$	$k=5$
LLR	$k=7$	$k=7$
SVMCK	$\mu=0.5$	$\mu=0.4$
JSR	$K_0=30$	$K_0=30$
PCSSR	$\lambda_1=0.01,\lambda_2=0.0004$	$\lambda_1=0.01,\lambda_2=0.0004$
SCSSR	$\lambda_1=0.001,\lambda_2=0.01$	$\lambda_1=0.001,\lambda_2=0.01$
BDLRR	$\lambda_1=0.1,\lambda_2=0.3,\lambda_3=25$	$\lambda_1=0.1,\lambda_2=0.3,\lambda_3=25$
SSBDCP	$\lambda_1=0.1,\lambda_2=0.3,\lambda_3=1,\lambda_4=125$	$\lambda_1=0.1,\lambda_2=0.3,\lambda_3=1,\lambda_4=125$
SSBDFCP	$\lambda_1=0.1,\lambda_2=0.06,\lambda_3=1,\lambda_4=125$	$\lambda_1=0.1,\lambda_2=0.01,\lambda_3=1,\lambda_4=125$

D. Analysis of Tradeoff Parameters

SSBDFCP has several tradeoff parameters, such as λ_1, λ_2, λ_3, and λ_4, which must be adjusted to balance the contributions of the FCP, block-diagonal representation, and spatial regularization. The parameters are divided into three groups in the experiment, and one parameter was changed, while the others were fixed. Fig. 7 shows the variations in OAs with respect to the tradeoff parameters. Notably, the performance of the SSBDFCP method is robust to noise, and λ_4 can be set in the range of [25, 125], which is empirically determined. Fig. 7(a) and (d) shows that the performance of the proposed method is generally sensitive to variations in λ_1 and λ_2. The overall precision of the method can be improved by adjusting λ_1 and λ_2 in the range of [0.06, 0.1] for Indian Pines and in the range of [0.01, 0.1] for PaviaU. Specifically, the performance is promising when λ_1 is not overly large or small and balances the extraclass representation. The parameter λ_2 is typically small to balance the latent subspace structure representation, as this representation can enforce the sparse property and produce the optimal result. Similarly, the overall precision of the proposed method can be improved by adjusting the parameter $\lambda_3 = 1$ for Indian Pines and PaviaU, as shown in Fig. 7(b) and (c) and (e) and (f), respectively. This finding indicates that the spatial and spectral information in an HSI plays an important role in feature extraction. The optimal parameters of each graph construction method are detailed in Table VI.

E. Analysis of the Block Structure Representation

For clarity, the binarized visual block-diagonal representation of the BDLRR and SSBDFCP methods is obtained for the PaviaU data set and is shown as block-diagonal representations of six classes in Fig. 8. Specifically, Fig. 8 shows that the proposed method can more clearly provide a block representation with true neighbors and yields better recognition results. Similarly, the SSBDFCP method promotes the low-rank property of intraclass samples and the sparsity of extra-class samples, which improves the compactness of the intraclass representation. Moreover, removing noise in the surrounding HSI pixels via the feedback of weak supervised information is a better option.



(a) (b)

Fig. 8. Block-diagonal representation comparisons of the BDLRR and SBDFCP methods for the PaviaU data set. The threshold value of the block-diagonal structure representation is 3e−3. (a) Binarized BDLRR. (b) Binarized SSBDFCP.

F. Run Time Analysis

Table VII shows that the run times of GK and LLR are faster than those of the other methods for the two data sets,

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TABLE VII
COMPUTATIONAL TIMES OF DIFFERENT METHODS FOR THE HSI DATA SETS (SECONDS)

	GK	LLR	SVMCK	JSR	EMP	EMAP	EPF	PCSSR	SCSSR	BDLRR	SSBDCP	SSBDFCP
Indian Pines	7.36	20.94	929.70	145.53	1205.87	1255.85	29.23	6974.53	6.23e+4	739.49	3097.80	773.78
PaviaU	52.95	256.71	1449	629.21	1859.78	1920.73	49.15	5.69e+4	6.05e+5	8536.78	9810.01	5670.82

TABLE VIII
OA WITH DIFFERENT WINDOW SIZES FOR TWO HSI DATA SETS

Data	Window size	3×3	5×5	7×7	9×9	11×11	13×13
Indian Pines	SCSSR	68.07	65.54	64.03	62.85	62.19	61.78
	JSR	86.81	89.40	89.71	90.74	87.86	86.16
	SVMCK	90.06	94.79	97.01	97.61	98.43	98.63
	SSBDCP	91.58	95.63	97.14	98.50	98.65	98.42
	SSBDFCP	**93.28**	**96.78**	**97.66**	**98.74**	**99.09**	**98.99**
PaviaU	SCSSR	50.14	72.96	69.94	68.02	65.56	64.37
	JSR	95.77	99.22	99.34	99.08	98.56	97.98
	SVMCK	95.76	96.91	97.48	98.20	98.91	98.16
	SSBDCP	94.01	98.79	99.42	99.23	99.06	99.02
	SSBDFCP	**96.37**	**99.34**	**99.53**	**99.43**	**99.27**	**99.16**

mainly due to the associated computational complexity. Notably, the PCSSR, SCSSR, and SSBDCP methods require more time to achieve convergence because the class probability solution is difficult to obtain. The computational time of the SSBDFCP method is faster than that of the SSBDCP method because the FCP can be obtained quickly than the class probability, indicating that SSBDFCP can be applied to larger data sets and maintain high accuracy. The computational time of the SSBDFCP method is approximately equal to that of the BDLRR method for Indian Pines, but the OA is higher than that of the BDLRR method. Moreover, the computational time of the SSBDFCP method is shorter than that of the BDLRR method for the PaviaU data set, indicating that it achieves faster convergence and yields better recognition precision. Although the SSBDFCP requires the most calculation time, the classification result is significantly improved. In addition, with the improvements in computing capacity, the time problem will no longer be a major problem [17].

G. Analysis of Spatial Window Size

To analyze the classification performance of the SSBDFCP method with different spatial window sizes, the performance of the SCSSR, JSR, and SVMCK methods with different spatial window sizes is tested on two data sets. Each method is repeated five times to avoid the bias due to the random sampling. The window size is selected from the set $\{3, 5, 7, 9, 11, 13\}$, and the classification results are reported in Table VIII. For SVMCK, we select the mean spectral values of the square window as the spatial features and employ the weighted summation kernel to balance the spatial and spectral components [46]. The parameters of the SVMCK and JSR in the experiment are reported in Table VIII, indicating that window sizes of 13×13 and 9×9 are optimal for the SVMCK and JSR methods in Indian Pines, respectively, and window sizes of 11×11 and 7×7 are optimal for the SVMCK and JSR methods in PaviaU, respectively. The window size of the SCSSR is from [17], which is 3×3 for the two data sets.

As reported in Table VIII, with the increased spatial window size, the OAs of the SSBDFCP method are always better than those of the other methods. The SSBDFCP method maintains a peak value with respect to the 11×11 window size in the Indian Pines data and 7×7 window size in the PaviaU data, which is determined by different sizes of the land-cover type. The reason that the SSBDFCP method is better than the other methods in OAs is that a large window size contains more spatial information from the same land-cover type to reveal the unique intrinsic structure of HSI. Moreover, it is demonstrated that the SSBDFCP method can better make full use of more valuable spatial structure information under the same spatial window size. However, the SCSSR method usually performs well with a smaller window size, which is determined by its own characteristics [17].

H. Analysis of Labeled Information

To further demonstrate the performance of the SSBDFCP method, the popular methods, such as the EMP, EMAP, EPF, GK, LLR, BDLRR, and SSBDCP methods, are compared with the SSBDFCP method in terms of different percentages of labeled samples from the two data sets. The parameters settings of EMP, EMAP, and EPF are consistent with [3], [4], and [44].

For better presenting the superiority of SSBDFCP, the spatial window size of SSBDFCP is set to 11×11 for the Indian Pines data and 7×7 for the PaviaU data, according to the analysis of the spatial window size for the two data sets. As shown in Fig. 9, the OA of different methods exhibits an upward trend as the number of labeled samples selected from each class increases. The OAs of the SSBDFCP method with different percentages of labeled samples from the two data sets are higher than those of EMP, EMAP, EPF, and the

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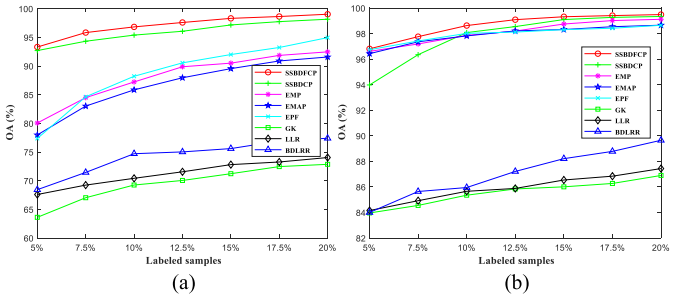


Fig. 9. OAs of different methods with different labeled samples for the two data sets. (a) Indian Pines. (b) PaviaU.

other methods. The main reason is that the FCP regularization explicitly reveals the "typicalness" of the latent subspace structure from the surrounding neighbors and provides feedback via "weakly" supervised information, with the assistance of the unique spatial information of HSI. However, the SSBDCP method does not perform well with 5% and 7.5% of labeled samples compared with the EMP, EMAP, and EPF methods because the class probability constraints obtained by SR are given equal importance in determining the class distribution of the sample to be classified. Thus, the SSBDFCP method is proposed to address the limitations of the SSBDCP method, and the FCP regularization can better reveal the "typicalness" of the latent subspace structure information based on the fuzzy class distribution of all samples.

VI. Conclusion

In this article, we propose the SSBDCP and SSBDFCP methods based on the block-diagonal structure representation of a latent subspace for HSI classification. First, the class probability is integrated into the SSBDCP method to explore the internal attribute features and enforce the sparsity constraint of BDLRR. Furthermore, the SSBDFCP method overcomes the limitation of the SSBDCP method with respect to lacking "typicalness" and has a shorter run time. Notably, the FCP is used to guide the graphical expression of the latent subspace structure of intraclass samples. Then, the relevant self-expressiveness vectors can be integrated to remove noise features around pixels and take advantage of the benefits of true neighbors. Finally, the latent relationship between the spectral and the spatial information is constructed to improve the spectral–spatial graph expression. The experimental results for the Indian Pines and PaviaU data sets show that the proposed methods achieve better recognition results than other popular graph construction methods.

The proposed methods still have some limitations. Specifically, achieving optimal results with these proposed methods is not easy because several parameters must be adjusted. In addition, the FCP, which requires the selection of adaptive neighborhoods, may yield poor classification results in some cases. In the future, a convolutional neural network, such as R-VCANet [47], can be incorporated into SSBDFCP to better express the spectral and spatial features of HSIs. Moreover, the more robust local covariance matrix representation (LCMR) [48] feature extraction technique and the SSBDFCP technique will be combined into a unified framework to explore more robust local and global features for HSI classification.

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